

Assignment Number: 3.5

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BATCH: 24

Question 1: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year.

Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions.

```
148 #write a python program to check if the given year is leap year or not taking user input
149 year = int(input("Enter a year: "))
150 ✓ if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
151     print(year, "is a leap year.")
152 ✓ else:
153     print(year, "is not a leap year.")
154 #user input is only positive integer handle negative and invalid inputs
155 ✓ try:
156     year = int(input("Enter a year: "))
157     if year < 0:
158         print("Please enter a positive integer for the year.")
159     else:
160         if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
161             print(year, "is a leap year.")
162         else:
163             print(year, "is not a leap year.")
164 ✓ except ValueError:
165     print("Invalid input. Please enter a valid integer for the year.")
166
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
.py
Enter a year: 2023
2023 is not a leap year.
Enter a year: 2000
2000 is a leap year.
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> |
```

Question 2: One-Shot Prompting (GCD of Two Numbers)

Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6

Task:

- Compare with a zero-shot solution.
- Analyze algorithm efficiency.

```
169 #write a well-commented python program to find GCD of two numbers
170 #example input 12, 18 output 6
171 def compute_gcd(a, b):
172     # Ensure a and b are non-negative integers
173     a = abs(a)
174     b = abs(b)
175     while b:
176         a, b = b, a % b
177     return a
178 try:
179     num1 = int(input("Enter first number: "))
180     num2 = int(input("Enter second number: "))
181     if num1 < 0 or num2 < 0:
182         print("Please enter non-negative integers only.")
183     else:
184         gcd = compute_gcd(num1, num2)
185         print("The GCD of", num1, "and", num2, "is:", gcd)
186 except ValueError:
187     print("Invalid input. Please enter valid integers.")
188
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS + v

```
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter first number: 24
Enter second number: 12
The GCD of 24 and 12 is: 12
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> 
```

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

- Input: 4, 6 → Output: 12
- Input: 5, 10 → Output: 10
- Input: 7, 3 → Output: 21

Task:

- Examine how examples guide formula selection.
- Test edge cases

```
191 #generate a well-commented python program to find LCM of two numbers
192 #example input 4, 6 output 12
193 #input: 5, 10 output: 10
194 #ex input 7, 3 output 21
195 def compute_gcd(a, b):
196     # Function to compute GCD using Euclidean algorithm
197     a = abs(a)
198     b = abs(b)
199     while b:
200         a, b = b, a % b
201     return a
202 def compute_lcm(a, b):
203     # Function to compute LCM using GCD
204     if a == 0 or b == 0:
205         return 0
206     gcd = compute_gcd(a, b)
207     lcm = abs(a * b) // gcd
208     return lcm
209 try:
210     num1 = int(input("Enter first number: "))
211     num2 = int(input("Enter second number: "))
212     if num1 < 0 or num2 < 0:
213         print("Please enter non-negative integers only.")
214     else:
215         lcm = compute_lcm(num1, num2)
216         print("The LCM of", num1, "and", num2, "is:", lcm)
217 except ValueError:
218     print("Invalid input. Please enter valid integers.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
.py
Enter first number: 7
Enter second number: 3
The LCM of 7 and 3 is: 21
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB>
```

Question 4: Zero-Shot Prompting (Binary to Decimal Conversion)

Write a zero-shot prompt to generate a Python function that converts a binary number to decimal.

Task:

- Test with valid and invalid binary inputs.
- Identify missing validation logic

```
222 #write a well-commented python program to convert binary to decimal and handle invalid inputs
223 def binary_to_decimal(binary_str):
224     decimal = 0
225     power = 0
226     for digit in reversed(binary_str):
227         if digit == '1':
228             decimal += 2 ** power
229         elif digit != '0':
230             raise ValueError("Invalid binary digit")
231         power += 1
232     return decimal
233 #take user input and test with valid and invalid inputs
234 if __name__ == "__main__":
235     binary_input = input("Enter a binary number: ")
236
237     if all(char in '01' for char in binary_input):
238         decimal_value = binary_to_decimal(binary_input)
239         print("The decimal value of binary", binary_input, "is:", decimal_value)
240     else:
241         print("Invalid input. Please enter a valid binary number.")
242
243
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
13.exe c:/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter a binary number: 345
Invalid input. Please enter a valid binary number.
osoft/windowsApps/python3.13.exe c:/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter a binary number: 1010
The decimal value of binary 1010 is: 10
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB>
```

Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

- Compare clarity with zero-shot output.
- Analyze handling of zero and negative numbers.

```
244 #write a well-commented python program to convert decimal to binary
245 #example input 10 output 1010
246 def decimal_to_binary(n):
247     if n < 0:
248         raise ValueError("Input must be a non-negative integer.")
249     binary_str = ""
250     if n == 0:
251         return "0"
252     while n > 0:
253         binary_str = str(n % 2) + binary_str
254         n //= 2
255     return binary_str
256 try:
257     decimal_input = int(input("Enter a decimal number: "))
258     if decimal_input < 0:
259         print("Please enter a non-negative integer.")
260     else:
261         binary_output = decimal_to_binary(decimal_input)
262         print("The binary representation of", decimal_input, "is:", binary_output)
263 except ValueError:
264     print("Invalid input. Please enter a valid non-negative integer.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/WindowsApps/python3.
/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter a decimal number: 1243454
The binary representation of 1243454 is: 100101111100100111110
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/WindowsApps/python3.
/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter a decimal number: r536
Invalid input. Please enter a valid non-negative integer.
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> 
```

Question 6: Few-Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks whether a number is a Harshad (Niven) number.

Examples:

- Input: 18 → Output: Harshad Number
- Input: 21 → Output: Harshad Number
- Input: 19 → Output: Not a Harshad Number

Task:

- Test boundary conditions.
- Evaluate robustness

```
269 # generate a well-commented python program to check whether a given number is a harshad number or not
270 def is_harshad_number(n):
271     if n <= 0:
272         raise ValueError("Input must be a positive integer.")
273     digit_sum = sum(int(digit) for digit in str(n))
274     return n % digit_sum == 0
275 try:
276     number = int(input("Enter a number: "))
277     if number <= 0:
278         print("Please enter a positive integer.")
279     else:
280         if is_harshad_number(number):
281             print(number, "is a Harshad number.")
282         else:
283             print(number, "is not a Harshad number.")
284 except ValueError:
285     print("Invalid input. Please enter a valid positive integer.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS + v

```
Enter a number: 21
21 is a Harshad number.
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:
/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter a number: 19
19 is not a Harshad number.
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:
/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter a number: 35
35 is not a Harshad number.
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> 
```